

Jahangirnagar University



Assignment On

Software Quality Factors (McCall's Quality Model)

Course Title: Software Project Management and Quality Assurance
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SUBMITTED TO

Dr. Mohammad Zahidur Rahman
Professor, Department of Computer Science and
Engineering

SUBMITTED BY

Md. Habibur Rahman
Id No: CSE202601013
Department of Computer Science and Engineering

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Part - B Analytical questions

1. Compare the three categories of McCall's Software quality factors using suitable examples?

Answer: Product operation (Running): Focuses on the immediate user experience and system stability.

Example: Ensuring a banking portal's liveness check algorithm executes quickly (Efficiency) and securely (Integrity).

Product revision (Adapting): Focuses on developers interaction with the codebase over time.

Example: Structuring the code cleanly so that adding a new "passport verification" module with a minimal effort (Flexibility).

Product transition (Moving): Focuses on the system's adaptability to new infrastructure.

Example: Reusing a previously built ORR component in a completely new loan-processing application (reusability).

2. Explain how poor software requirements can negatively affect software quality?

Answer: If non-financial requirements are ignored during the initial software requirement specification (SRS), resulting architecture will be flawed. For instance, if data load (Efficiency) is not specified, developers might build a system that works perfectly for 10 users but crashes for 10,000. This leads to massive technical debt, severe operational downtime and costly code rewrites.

3. A hospital information system requires 99.999% availability. Which McCall quality factors are most critical? Justify your answer.

Answer: Reliability: The system must not crash as downtime could literally risk patient lives.

Correctness: Medical data must be precisely accurate, incorrect dosage or patient records are catastrophic.

Integrity: patient health records are highly sensitive and require strict privacy controls to prevent unauthorized access.

4. Distinction between Reliability, safety and survivability?

Answer: Reliability: Focuses on the frequency of failure (e.g., mean time between failure)

safety (Deutsch & Willis): Focuses on preventing the system from causing hazards or damage when it does fail. (e.g., tagging a fallback alarm)

survivability (Deutsch & Willis): Focuses on business continuity and the speed of recovery after a failure occurs (e.g., mean time to recover).

5. Explain why Reusability can reduce software development cost & time?

Answer: By designing software modular, decoupled blocks (like microservices), components can be dropped into new projects without starting from scratch. Reusing a heavily tested, production ready module, eliminates the need to spend billable hours on redesigning, recording and retesting that specific functionality.

Part - C

Scenario-Based Questions

1. A banking application must:
- prevent unauthorized access
 - process transactions within 2 seconds
 - Operate continuously with minimal downtime.

Answer to scenario: Banking Application

Based on McCall's software quality model, the relevant quality factors are:

1. prevent unauthorized access → Integrity.
- Why it is important: It ensures that the system

use robust security to protect sensitive user data prevent cyber attacks and stop financial fraud.

2. process transactions within 2 seconds

→ Efficiency

why it is important: It ensures the software optimally utilize hardware resources so that transactions are processed initially lagging or not during heavy traffic.

3. Operate continuously with minimal downtime → Reliability.

why it is important: Banking requires 24/7 availability, high reliability ensures the system is fault tolerant.

3. E-commerce company scenario

context: Run on Windows, Linux, Android, cloud and integrate with payment gateway.

portability: The application must operate seamlessly across entirely different operating system and hardware architecture.

Inter Operability: The platform can not process sales without talking to external, third party payment gate way.

Justification: without portability the multi platform strategy fails. without inter operability the core business functions fails.

4. A software company frequently modifies its product for different client. which product revision Factors are most important? Explain with examples.

Answer:

modifying products for different clients

Flexibility: The most critical factors. The code base must be designed so that custom feature can be toggled or added for client A without breaking the core product for client B.

Maintainability: frequent modifications increase the risk of bugs; The code must be clean and well-documented so that developers can quickly trace and resolve client specific errors.

5. compare McCall's model with alternative models proposed by Evans & Marciniak and Deutsch & Willis. Which additional quality factor do you consider most valuable in modern software engineering? Explain.

Answer: McCall's vs Alternative Models is a foundational model lacks considerations for modern operational tracking and scaling.

Evans & Marciniak: Added Verifiability and Expandability.

Deutsch & Willis: Added Safety, Manageability and Survivability.

most valuable today: Expandability (scalability). In the era of cloud computing and microservices, an application ability to dynamically scale hardware resources up or down based on traffic spikes is a mandatory requirement for enterprise software.

Part - D : Higher - Order Thinking

1. Is McCall's model still relevant for AI-based and cloud-native system?

Answer: yes, it acts as a strong foundation but it requires modernization.

Relevance: core factors like reliability, integrity and interoperability are just like vital today.

Limitations: McCall's does not account for cloud specific traits like Elasticity. Furthermore for AI systems, correctness becomes ambiguous.

2. Design a quality requirements specifications for an online university learning management system using McCall's quality factors.

Answer: Operation: Intuitive UI using accessible front-end templates so non technical students can easily navigate courses instantly.

Revision: Automated CI/CD pipeline must verify that new quiz features do not break

existing gradebook calculations.

Transition: (Interoperability) must sync grades directly to universities central administrative database and integrate with external SSO.

3. Explain how software quality factors influence SRS? Discuss with example.

Answer: Quality factors transform a SRS from a simple "wishlist" into an actionable architecture blueprint.

Example: If the SRS defines high flexibility and reusability, the software team will likely choose a microservices architecture using Docker and Node.js rather than a monolithic architecture.

4. If you were designing an autonomous vehicle control system, which quality factors would you prioritize and why?

Answer: Autonomous vehicle control system properties.

safety: one absolute highest priority.

Reliability (McCall): The software can't crash. While driving at 20 mph, the MTBF must be practically infinite.

Viva/Discussion Question

1. Which software quality factor is the most difficult to measure? why?

Usability and maintainability: They are highly subjective and depend on human factors (e.g., how intuitive a user finds an interface or how easily a developer understands someone else's code).

2. Can software be highly efficient but have poor usability? Explain.

Yes, A backend system might process complex algorithms in milliseconds but output the result via a command-line interface that requires complex syntax to operate.

3. Why should clients be involved in defining software quality requirements?

Answer: clients understand the business context. They know what degree of downtime will cost them money or what level of security their industry regulators demand.

4. Are quality factors equally important for every software project? justify your answer.

Answer: NO. A mobile game prioritizes usability and efficiency (graphics rendering) whereas a banking portal prioritizes integrity (security) and reliability over flashy UI.

5. Which quality factor is most critical for AI-powered healthcare system?

Answer: Safety and correctness: An incorrect AI diagnosis or a system failure during robotic surgery directly threatens human life.